Number Representation: Decimal to Binary and Hexadecimal

Data Representation & Binary Number Systems • Standard: CCSS.MATH.PRACTICE.MP5

1. MATHEMATICAL CONCEPT & STACK APPROACH: Represent integers physically as [0] and [1] bit registers on the Place Value Tray and computationally convert between bases with [f:] [BASE].

Binary (base-2) and hexadecimal (base-16) follow the exact same positional place-value principle as base-10, using powers of 2 (8, 4, 2, 1) instead of powers of 10. Switching base modes on the stack changes how the active value in Register X is grouped and displayed without changing the underlying number.

2. MODELING WITH BLOCKS (WHEN APPLICABLE) & STUDENT PREDICTION

Published Block Kit: Place Value Studio (Binary [0]/[1] Register Trays & Base-N Columns)

Hands-On Setup: Use the sockets of the Place Value Tray as binary bit positions (8, 4, 2, 1). Place [1] Digit Blocks for active '1' bits and [0] Digit Blocks for '0' bits.

Student Prediction (Before Calculator): Place [1], [0], [1], [1] in the 8, 4, 2, 1 sockets: binary $1011_2 = 8 + 0 + 2 + 1 = 11_10$. Predict hex value: 0B_16.

3. CALCULATOR EXECUTION (SC-32 KEYCAPS & 4-LEVEL STACK TRACE)

Keystroke Flow: [1] [1] [↶] [BASE]

Key Pressed	Screen (X)	Internal 4-Level Stack Effect (T, Z, Y, X)
[1] [1]	`11`	Enter decimal integer 11 in DEC mode
[↶] [BASE]	`HEX 0B`	Press [f:] [BASE]: screen displays Hexadecimal 0B!
[↶] [BASE]	`BIN 1011`	Press [f:] [BASE]: screen displays Binary 1011!
[↶] [BASE]	`OCT 13`	Press [f:] [BASE]: screen displays Octal 13!
[↶] [BASE]	`DEC 11`	Press [f:] [BASE]: returns to Decimal 11

4. STORY CONTEXT (Leibniz's Binary Medallion, Claude Shannon's 1937 Relay Basement & NASA's 36-Bit IBM 7090 Word)

In 1697, Gottfried Wilhelm Leibniz designed a medallion showing that every number can be built from 1 and 0 alone; in 1937, 21-year-old MIT student Claude Shannon proved electrical switches execute Boolean algebra; and in 1961, NASA's IBM 7090 computed orbital trajectories in 36-bit binary words! Using [0] and [1] Digit Blocks on the Place Value Tray and [BASE], students decode bytes, hex nibbles, and bitwise masks.

Core Investigation (Student Warm-Up A):

Enter decimal 255. Use [f:] [BASE] to convert to HEX and BIN. What is the binary pattern?

Bitwise Logic: AND Masking & Network Subnets

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1. EXTENSION CONCEPT (WARM-UP B), BLOCKS & PREDICTION

Concept & Stack Approach: Bitwise logic operations (AND, OR, XOR, NOT) compare corresponding binary place-value columns of two integers in parallel. Placing value A on Waiting Shelf Y and mask B on Workbench X lets [AND] filter bit columns in a single stack operation.

Blocks & Prediction: Row 1 of the Place Value Tray holds nibble A ([1][1][0][0] = 12). Row 0 holds mask B ([1][0][1][0] = 10). Prediction: Bitwise AND: bit 3 (1 & 1 = 1), bit 2 (1 & 0 = 0), bit 1 (0 & 1 = 0), bit 0 (0 & 0 = 0). Result: 1000_2 = 8_10.

2. EXTENSION CALCULATOR SEQUENCE & STACK TRACE

1 2 ENTER 1 0 AND

Key Pressed	Screen (X)	Internal 4-Level Stack Effect
1 2	12	Enter operand A = 12 (binary 1100)
ENTER	12	12 lifted to register Y
1 0	10	Enter operand B = 10 (binary 1010) in X
AND	8	Bitwise AND evaluates 1100 & 1010 = 1000 (8 decimal)!

3. TARGET MISCONCEPTIONS & HANDS-ON BLOCK REMEDIATIONS

Misconception 1: Confusing bitwise [AND] / [OR] with decimal addition or multiplication (e.g., thinking 12 AND 10 is 22).

Hands-On Block Remediation: Model 12 ([1][1][0][0]_2) in Row 1 and 10 ([1][0][1][0]_2) in Row 0 of the Place Value Tray using [1] and [0] Digit Blocks; only columns where BOTH rows hold [1] remain 1 in 12 AND 10 = 8 ([1][0][0][0]_2).

Misconception 2: Reading hexadecimal [10] as ten instead of decimal 16 ($1 \times 16^1 + 0 \times 16\text{^}0$).

Hands-On Block Remediation: Place [1] in the Sixteens column and [0] in the Ones column on the Place Value Tray, then switch from [HEX] [10] to [DEC] on the SC-32 to see 16.

4. EXTENSION MATHEMATICAL INSIGHT

Bitwise logic operations (AND, OR, XOR, NOT) compare corresponding binary place-value columns of two integers in parallel. Placing value A on Waiting Shelf Y and mask B on Workbench X lets [AND] filter bit columns in a single stack operation.

Digital Logic, Base Conversions & Computer Architecture

Name: _____ Date: _____ Class: _____

STUDENT REFERENCE GUIDE: NUMBER BASES & BITWISE LOGIC GATES

- Binary (Base 2): digits [0] and [1] on the Place Value Tray. Hexadecimal (Base 16): digits 0-9 and A-F.
- Press [f:] [BASE] to cycle between DEC (decimal), HEX (hexadecimal), OCT (octal), and BIN (binary).
- Bitwise AND: result bit is 1 only if BOTH Row 1 and Row 0 sockets hold [1] (used for masking and subnets).

Key Sequence: [1] [1] [↖] [BASE]

Investigation 1: Convert decimal 42 to binary using [f:] [BASE].

1. Geometric / Area / Coordinate Sketch:

2. Algebraic Setup & Stack Plan:

Algebraic Setup:

Stack Registers (Y, X):

RPN Keystroke Plan:

3. My Prediction:

Final Verified Answer:

[ ] Checked on SC-32

Investigation 2: Convert hexadecimal 2A to decimal.

1. Geometric / Area / Coordinate Sketch:

2. Algebraic Setup & Stack Plan:

Algebraic Setup:

Stack Registers (Y, X):

RPN Keystroke Plan:

3. My Prediction:

Final Verified Answer:

[ ] Checked on SC-32

Investigation 3: Evaluate bitwise AND: 15 (1111) AND 7 (0111).

1. Geometric / Area / Coordinate Sketch:

2. Algebraic Setup & Stack Plan:

Algebraic Setup:

Stack Registers (Y, X):

RPN Keystroke Plan:

3. My Prediction:

Final Verified Answer:

[ ] Checked on SC-32

Investigation 4: Evaluate bitwise XOR: 12 (1100) XOR 10 (1010).

1. Geometric / Area / Coordinate Sketch:

2. Algebraic Setup & Stack Plan:

Algebraic Setup:

Stack Registers (Y, X):

RPN Keystroke Plan:

3. My Prediction:

Final Verified Answer:

[ ] Checked on SC-32

Unit 12: Digital Logic, Base Conversions & Computer Architecture — Answer Key

Complete solutions and SC-32 verification keystrokes for Grades 9–12 / AP CS

1. CORE (WARM-UP A) & EXTENSION (WARM-UP B) SOLUTIONS

Core Investigation (Warm-Up A): Enter decimal 255. Use [f:] [BASE] to convert to HEX and BIN. What is the binary pattern?

Core Solution: HEX: FF, BIN: 11111111 (8 ones). Keystrokes: [2][5][5] [f:][BASE]

Core Check Keys: 2 5 5 ↰ BASE

Extension Challenge (Warm-Up B): Calculate bitwise OR for 12 (1100) and 3 (0011). Predict the binary and decimal result.

Extension Solution: Binary: 1111, Decimal: 15. Keystrokes: [1][2] [ENTER] [3] [OR]

Ext. Check Keys: 1 2 ENTER 3 OR

2. STUDENT LAB WORKSHEET SOLUTIONS (INVESTIGATIONS 1–4)

Investigation 1: Convert decimal 42 to binary using [f:] [BASE].

Expected Answer: 101010 (32 + 8 + 2)

SC-32 Verification Keys: 4 2 ↰ BASE

Investigation 2: Convert hexadecimal 2A to decimal.

Expected Answer: 42

SC-32 Verification Keys: In HEX mode: 2 A -> switch to DEC

Investigation 3: Evaluate bitwise AND: 15 (1111) AND 7 (0111).

Expected Answer: 7 (0111)

SC-32 Verification Keys: 1 5 ENTER 7 AND

Investigation 4: Evaluate bitwise XOR: 12 (1100) XOR 10 (1010).

Expected Answer: 6 (0110)

SC-32 Verification Keys: 1 2 ENTER 1 0 XOR

Hands-On Block & Stack Verification Protocol

Students first build the physical block model and write their mathematical strategy and prediction before verifying their result on the StackCalc calculator.